Title	Authors	Abstract	EXTRA
How do disparate urbanization and climate change imprint on urban thermal variations? A comparison between two dynamic cities in Southeast Asia	Can Trong Nguyen, Amnat Chidthaisong , Atsamon Limsakul, Pariwate Varnakovida, Chaiwat Ekkawatpanit Phan Kieu Diem , Nguyen Thi Hong Diep	**Abstract:** This study assessed the combined influences of climate change and urbanization on land surface temperature (LST) and surface urban heat island (SUHI) in the Bangkok metropolitan region (BKK) and Ho Chi Minh City metropolitan area (HCM) during 1990–2020. We found climate change evidence from trends and variations of temperature extremes in both cities throughout the past three decades, especially for the significant upward trends of nighttime temperature in HCM (0.23 – 0.3 °C/decade). These cities have been undergoing rapid urbanization, in which HCM has a higher urbanization rate with an annual growth rate of 8.03%/year against BKK (4.26%/year). Landscape metrics have rapidly shifted towards urban aggregation and green space fragmentation along with urbanization. A peer comparison shows that BKK holds a greater urban size and LST intensity, while HCM leads in urban growth and SUHI magnitude. These changes jointly contribute to climbing SUHI intensity, especially the dramatic influence of surface changes. Yet, temperature extremes also play an indispensable role in regulating SUHI magnitude, in which the impact in HCM is considerable compared to the effect in BKK. The research findings provide crucial information for both cities – where incorporation of climate change resilience strategies and SUHI mitigation into the master plans for livable cities are recommended.	
Thermal intensification and urban land cover change in Southeast Asian capitals: over three decades of satellite observations and trend analysis	Yong Le Ng, Ming Han Lim, Yuk Feng Huang & Yee Ling Lee	**Abstract:** Urbanisation in Southeast Asia has intensified land surface warming and Intra-Urban Heat Island Intensity (IUHII) effects. This study examined Land Surface Temperature (LST), IUHII, and Land Use/Land Cover (LULC) changes across eleven Southeast Asian capital cities from 1990 to 2024. LST retrieval employed the radiative transfer equation, while LULC was classified via a Random Forest algorithm with above 97% accuracy. IUHII was calculated from urban–green space temperature differences. Significant warming trends emerged, especially in Vientiane (0.350 °C/year), Kuala Lumpur (0.235 °C/year), and Bangkok (0.215 °C/year). LULC classification revealed major urban expansion with vegetation and water loss. This first multi-decadal, cross-city IUHII assessment in tropical Southeast Asia addresses a key research gap. However, this study is limited by its focus on capital cities, reliance on historical trends, and exclusion of socio-economic and governance factors. Nevertheless, the results provide critical empirical evidence to guide climate-resilient urban planning and support green infrastructure integration.	

Cooling the cities: A comprehensive review of urban heat island mitigation strategies in Southeast Asia	Kyaw Zabu Tun, Malay Pramanik, Suraj Kumar Mallick, Rabin Chakrabortty, Bijay Halder, Kanak N. Moharir, Chaitanya Baliram Pande, Mohamed Zhran	**Abstract:** This systematic review synthesizes 27 peer-reviewed studies (2020–2023) on urban heat island (UHI) mitigation in Southeast Asia, focusing on cooling effectiveness and implementation feasibility. In dense cities like Singapore and Kuala Lumpur, green infrastructure—urban forests, green roofs, and street-tree networks—proves most effective, reducing air temperatures by up to 2.6 °C and surface temperatures by 11 °C. Engineered solutions, such as high-albedo coatings and permeable pavements, offer 4–9 °C of surface cooling and added benefits like stormwater control, though tropical humidity, glare, and maintenance pose challenges. Informal settlements demonstrate the value of low-cost, community-led cooling efforts using shade structures and vegetation. Despite these benefits, barriers persist: high costs, limited urban land, and fragmented governance systems hamper the scalability of hybrid green-grey solutions. Evaporation-based strategies are further constrained by complex urban geometries and monsoonal climates. Fewer than 20 % of Southeast Asian cities incorporate UHI mitigation into formal planning frameworks, limiting systemic uptake. To advance climate-resilient urbanism, interdisciplinary collaboration is needed. Future priorities should include long-term monitoring in tropical climates, integrated trials of hybrid interventions, and cost–benefit models that capture health, energy, and flood resilience co-benefits. Embedding these approaches into planning systems, as seen in Bangkok’s zoning reforms and Singapore’s Green Mark, is key to sustainable adaptation.	

Spatiotemporal Analysis Of Urban Land Surface Temperature (Ulst) In Davao City	Y. Zha,J. Gao &S. Ni	**Abstract:** Remotely sensed imagery is ideally used to monitor and detect land cover changes that occur frequently in urban and peri-urban areas as a consequence of incessant urbanization. It is a lengthy process to convert satellite imagery into land cover map using the existing methods of manual interpretation and parametric image classification digitally. In this paper we propose a new method based on Normalized Difference Built-up Index (NDBI) to automate the process of mapping built-up areas. It takes advantage of the unique spectral response of built-up areas and other land covers. Built-up areas are effectively mapped through arithmetic manipulation of re-coded Normalized Difference Vegetation Index (NDVI) and NDBI images derived from TM imagery. The devised NDBI method was applied to map urban land in the city of Nanjing, eastern China. The mapped results at an accuracy of 92.6% indicate that it can be used to fulfil the mapping objective reliably. Compared with the maximum likelihood classification method, the proposed NDBI is able to serve as a worthwhile alternative for quickly and objectively mapping built-up areas.	
Modeling the influence of land cover dynamics on spatio-temporal variations in land surface temperature in Cagayan de Oro River basin, Mindanao, Philippines	Rylle Adriane Galvez, Hilly Ann Roa-Quiaoit, Frandel Louis Dagoc, Jaime Guihawan & Peter Suson	**Abstract:** The onset of agricultural intensification and urban expansion in Cagayan de Oro River basin has been continuously changing its landscape over the years. While it undergoes modifications its land cover, its associated impact on the land surface temperature over the years has not been evaluated. This study aims to model the spatio-temporal dynamics of land cover (2003–2020) and determine its influence on the land surface temperature variations in Cagayan de Oro River basin, Mindanao, Philippines from 2003 to 2020 using geographical information systems (GIS) and remote sensing techniques. Significant changes were observed in annual and perennial croplands which experienced substantial increases and is preceded by built-up areas. Considerable decreases were however recorded in shrubland, grassland, mangroves, and open forest. Between 2003 and 2020, the mean land surface temperature in Cagayan de Oro River basin increased by 5.48 ℃. The results indicate that built-up areas and bareland consistently recorded the highest mean LST values, while water bodies, mangroves, and forest cover had the lowest temperatures. This study promotes evidence-based decision-making for sustainable land use planning and management to mitigate the increasing land surface temperature in Cagayan de Oro River basin.	

When the city heats up: mapping urban heat risks through environmental and socioeconomic factors in Quezon City, Philippines	Aerol Cedrick Treyes	A**bstract:** While urbanization drives economic growth and development, it also intensifies urban heat, worsened by climate change and urban heat island effects, which continue to threaten public health, livability, and urban resilience in cities. This study develops and maps a localized heat risk index (HRI) for Quezon City, Philippines, integrating environmental exposure and socioeconomic vulnerability variables through a weighted framework derived using analytic hierarchy process based on expert input. Spatial analysis revealed that 81% of barangays, accounting for 70% of the city's population, fall under high-risk HRI classification (0.61–0.80), with the highest mean HRI recorded in Barangay Manresa (0.731). High-risk areas are concentrated in the southern and southwestern zones of the city, characterized by dense urbanization, limited vegetation, and high surface temperatures. Intraurban heat islets, covering 31% of the study area, strongly overlap with high-risk HRI barangays, emphasizing the compounded effects of environmental and socioeconomic factors on heat risk. The study provides insights to complement the Quezon City Enhanced Local Climate Change Action Plan 2021–2050 by identifying priority areas for implementing nature-based solutions, such as expanding green corridors and enhancing urban biodiversity. These findings highlight the critical role of integrating vegetation, reflective surfaces, and reduced built-up densities in mitigating heat risks and improving urban resilience. The HRI framework offers a replicable model for assessing urban heat risks, supporting evidence-based policymaking, and enhancing adaptive capacity in other rapidly urbanizing cities	

The effect of urbanization on temperature indices in the Philippine	John A. Manalo, Jun Matsumoto, Hiroshi G. Takahashi, Marcelino Q. Villafuerte II, Lyndon Mark P. Olaguera, Guoyu Ren, and Thelma A. Cinco	**Abstract:** This paper presents a comprehensive analysis of the effect of urbanization on the surface air temperature (SAT) from 1951 to 2018 in the Philippines. The daily minimum temperature (Tmin) and daily maximum temperature (Tmax) records from 34 meteorological stations were used to derive extreme temperature indices. These stations were then classified as urban or rural based on satellite night-lights. The results showed a significant difference in the SAT trends between urban and rural stations, indicative of the effect of urbanization in the country. Larger and more significant warming trends were observed in indices related to Tmin than those related to Tmax. In particular, the effects of urbanization were significant in the annual index series of Tmin, diurnal temperature range, minimum Tmin, percentage of days when Tmin was less than the 10th percentile (TN10p), percentage of days when Tmin was greater than 90th percentile (TN90p), and the number of coldest nights. The effects of urbanization were not as clear on the index series of maximum Tmax (TXx), minimum Tmax (TXn), percentage of days when Tmax was less than 10th percentile (TX10p), and the number of hottest days. The effects of urbanization on the annual series of extreme temperature indices were statistically significant at the 95% confidence level, with the exception of Tmax, TXn, TXx, TX10p, and the number of hottest days. Further analysis revealed that the effect of urbanization was the greatest during the DJF (December–January–February) season. These findings serve as a	**Conclusions:** The Philippines experiences several climatic hazards and environmental challenges as a result of climate change and urbanization. As such, it is essential to obtain an indepth understanding of extreme climate change and the potential impact of urbanization on observed climatic change. Cinco et al. (2014) analysed temperature trends in the Philippines, suggesting that days and nights were becoming warmer. However, their findings were influenced by several factors, such as urbanization, which plays an important role in observed temperature trends. Findings: Nights are warming faster than days: Urbanization has significantly increased minimum (nighttime) temperatures (T_{min}) in the Philippines, while its effect on maximum (daytime) temperatures (T_{max}) is much weaker. **Extremes are shifting:** Due to urbanization, there is a significant decrease in the number of cold nights and a corresponding increase in warm nights. **Shrinking temperature gap:** Because nighttime temperatures are rising faster than daytime temperatures, the diurnal temperature range (the difference between the daily high and low) is significantly decreasing. **Seasonal peak:** The warming effect caused by urbanization is most intense during the cool and dry season (December–January–February).

GIS-Based Spatio-Temporal Analysis of Vegetation Dynamics, Urban Expansion, and Land Surface Temperature in Davao City	Richelyn C. DumdumJason Ben R. Paragamac	**Abstract:** The Earth's surface and atmosphere have been substantially modified by the growth of anthropogenic activities and urbanization. Changes in the composition and operation of the urban ecological system will affect the health of city dwellers by increasing the risk of heat exposure, increasing the risk of violence, illness, and mortality, and reducing labor productivity. Davao City is currently undergoing rapid urbanization, and no comprehensive study has examined the effects of the changes on vegetation indices, urban expansion, traffic density, and land surface temperature, which are likely to affect the area. This study analyzes the spatio-temporal change in vegetation indices, built-up expansion, traffic density, and land surface temperature from 2018 to 2022 using Landsat 8 imagery and GIS tools, particularly QGIS. The results show a moderate negative relationship between NDBI and NDVI ($r = -0.4313$), a strong positive relationship between NDBI and LST ($r = 0.6216$), and a weak negative relationship between NDVI and LST ($r = -0.2495$). Results highlight the interconnected influence of urbanization, vegetation dynamics, and thermal conditions within the study area, underscoring the need to integrate green infrastructure, particularly along major road networks, to mitigate rising land surface temperatures.	
Land-use/land cover change and land surface temperature in Metropolitan Manila, Philippines using landsat imagery	King Joshua Almadrones-Reyes & Nikki Heherson A. Dagamac	**Abstract:** Remote sensing technology has been widely used to address the issue of urbanization for monitoring. In Metropolitan Manila, Philippines, there is a decline in vegetation over the last decade. This leads to the excess heat in the surface temperature and a phenomenon known as Urban Heat Island. In this study, the land surface temperature (LST), land-use/land cover changes (LULC), normalized difference vegetation index (NDVI) and normalized difference built-up index (NDBI) in 2001 and 2019 were retrieved using Landsat 7 Enhanced Thematic Mapper + (ETM +). Our results showed the following: (i) an increase in the mean temperature of 4.02 °C from 2001 to 2019 in MM, (ii) rise in NDBI and a decline in NDVI values is correlated with an increase in LST over the last years in MM; (iii) LULC maps affirmed the highest occupancy of built-up areas caused by possible urban sprawl explaining the increase in warming in MM. This study provides added evidence that urbanization activities in Metro Manila leads to a warming effect that drives discomfort to many average Filipinos. Hence, environmental policies that would implement proper urban design utilizing more green spaces is recommended for local government units in MM to support.	

Urban Effects on Land Surface Temperature in Davao City, Philippines	Myla Ross Tinoy, A. U. Novero, K.P. C. Landicho, Alvin Baloloy, Ariel Blanco	**Abstract:** This study produced spatiotemporal hot and cold spot occurrence maps for Davao City for the period 1994-2019 using land surface temperature (LST) images. Urban heat is theorized to have been affected by some, if not all, of the following impact factors: air pollutant concentrations/particulate matter (PM10), vegetation “abundance” (using EVI), building “density” (NDBI), albedo, topography, and population density. A mobile traverse sampling was performed in the morning and afternoon of 15 April 2019 to measure PM10 in the city’s identified hot spots. The remaining factors were generated from imagery (i.e., Landsat 8, Synthetic Aperture Radar) and obtained from the Philippine Statistics Authority. These factors were analyzed against the LST which was obtained through Project GUHeat’s methodology. The relationships between the factors and LST were studied through multiple and quantile regression models (MRM & QRM). Results showed that variable PM10 does not have any significance in the MRM. Meanwhile, QRM were fitted to different quantile values, namely: 10th, 25th, 50th, 75th, and 90th. It is only at the 90th quantile where all the independent variables are good predictors for the LST. Albedo is the most important predictor for the LST at 10th quantile whereas Elev for the 25th quantile. However, when LST is at the 50th, 75th, and 90th quantiles NDBI is the most significant variable at predicting LST. Reliable spatiotemporal assessment and modelling of surface temperature are essential for urban planning and management to formulate sustainable strategies for the welfare of people and environment.	**Conclusions and Recommednations:** This study investigated the spatiotemporal distribution of LSTand its quantitative relationships with atmospheric,demographic, and remote sensing-derived parameters in DavaoCity. The results revealed that occupants of the main urbancenter (Districts of Poblacion, Talomo, Agdao, and Buhangin)and residents from coastal barangays of Bunawan and TorilDistricts experienced the warmest temperatures in Davao Cityfor the period 1994-2019. Multiple regression analysis showedthat the air pollutant variable (PM10) exhibited no significance,hence was dropped in the model. Albedo has greater impact inthe MRM followed by NDBI, Elev, Pop_Dens, and EVI. Quantile regression analysis further revealed that at the 90thquantile (31.87 °C – 33.42 °C) where all impact factors are goodpredictors, the presence of buildings in the city (NDBI) showedthe biggest impact for the LST. The result of these two regressionmethods indicated that the models are reliable enough to expressLST of Davao City, but also suggested that other impact factorsare more significant in order to effectively understand thevariations of LST. This study recommends that higher spatialand temporal resolution of satellite imageries should beemployed to continuously improve the HS and CS analyses.Moreover, further research should be done by replacing otherimpact factors in the model and try to include surface moisture,distance to water, solar insolation, and sky view factor. Lastly,to evaluate and compare the urban expansion of other developedareas in Mindanao such as Digos City, Panabo City, and TagumCity performing the same procedure of this study is suggested.

Assessment of Intra-Urban Heat Island in a Densely Populated City Using Remote Sensing: A Case Study for Manila City	Mark Angelo Purio, Tetsunobu Yoshitake, and Mengu Cho	**Abstract:** Changes in the environment occur in cities due to increased urbanization and population growth. Sustainable Development Goal (SDG) 11 is intrinsically linked to the environment, one facet of which is the need for universal access to secure, inclusive, and accessible green and public places. As urban heat islands (UHI) have the potential to negatively influence cities and their residents, existing resources and data must be used to identify and quantify these effects. To address this, we present the use of satellite-derived (2013–2022) and meteorological data (2014–2020) to assess intra-urban heat islands in Manila City, Philippines. The assessment includes (a) understanding the temporal variability of air temperature measurements and outdoor thermal comfort based on meteorological data, (b) comparative and correlative analysis between common Land-Use Land Cover indicators (Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI) and Normalized Difference Built-up Index (NDBI)) and Land Surface Temperature (LST), (c) spatial and temporal analysis of LST using spatial statistics techniques, and (d) generation of an intra-urban heat island (IUHI) map with a recommended class of action using a suitability analysis model. Finally, the areas that need intervention are compared to the affected population, and suggestions to enhance the thermal characteristics of the city and mitigate the effects of UHI are established.	**Conclusions:** This study presents the use of satellite-derived data and meteorological data to assess the presence of an intra-urban heat island in Manila City, Philippines. To address SDG 11 and provide better insights to make cities and human settlements inclusive, safe resilient, and sustainable in terms of UHI, different assessment methods were used and established. The assessment includes (a) understanding the temporal variability of air temperature measurements and outdoor thermal comfort based on meteorological data, (b) comparative and correlative analysis between common LULC indicators (NDVI, NDBI, and NDWI) to LST, (c) spatial and temporal analysis of LST using spatial statistics techniques, and (d) generation of an intra-urban heat island (IUHI) map with a recommended class of action using a suitability analysis model. Finally, the areas that need intervention are compared to the affected population, and suggestions to enhance the thermal characteristics of the city and mitigate the effects of UHI were established. Results show that there exists a clear difference between cold and warm areas within Manila City. Overall, residential areas, asphalted and concrete roads and walkways, and some commercial establishments and buildings exhibit higher surface temperatures compared to areas with vegetation and near bodies of water. Based on the results, mitigation strategies applicable to Manila City were proposed to improve the areas which need intervention.

Thermal changes along the urban-rural continuums in Southeast Asia	Yuanhui Zhu*, Soe W Myint, Jiquan Chen, Peilei Fan, Karen C Seto, Atul Jain, Jiaguo Qi and Jin Wang	**Abstract:** Southeast Asia's (SEA) rapid urbanization has intensified urban heat effects. While previous studies address urban warming at specific time points, the dynamic relationship between land cover and land use (LCLU) changes and land surface temperature (LST) along urban-rural continuums remains understudied. This study examines LST, LCLU, and NDVI changes across 19 SEA cities of varying urbanization levels from 2000 to 2022 using Landsat images. Results show that LST change rates differ by urbanization levels, with built-up lands driving urban warming. Larger cities exhibited more intense and widespread LST increases extending into rural areas. The megacities experienced significant LST growth within 5 km of city centers despite limited built-up expansion. However, some areas in many cities experience decreasing trends of heat. NDVI significantly influenced LST changes through its cooling effect and the warming impact due to vegetation loss caused by urban expansion. Findings support urban planning strategies that balance growth with enhancing green spaces, redirecting growth to smaller cities, and shifting development from city centers to urban outskirts.	

Integrating Satellite-Based and Participatory Spatial Analyses for Local, Contextualised, and Nuanced Extreme Heat Hotspot Mapping in Overheated Urban Southeast Asia	aurence L. Delina, Homer Pagkalinawan, Sharon Feliza Ann Macagba	**Abstract:** Rapid urbanisation in Southeast Asia intensifies extreme-heat exposure, especially where high population density, extensive impervious surfaces, and limited green cover coincide. This study combines multi-decadal satellite-derived land surface temperature (LST) mapping with spatial regression and community participatory mapping to locate community-level heat clusters of extreme heat in Metro Manila and the Bangkok Metropolitan Area. Using a 1 km grid and a suite of built-environment, infrastructural, natural feature, and socioeconomic covariates (including two policy-oriented efficiency ratios), we find that: (1) LST clusters exhibit significant positive spatial autocorrelation, and (2) a statistically significant quadratic relationship links a building-efficiency ratio to local temperatures in both cities—implying an optimal level of compactness beyond which further densification reduces surface heat (consistent with increased shading and compact-form benefits). In addition, higher population density, dense road networks, and proximity to extensive heat-retaining infrastructure (notably airports) are consistently associated with elevated temperatures, whereas greater greenspace accessibility is associated with cooling. Crucially, participatory mapping supplied fine-scale, lived-experience data that (a) identified microhotspots that LST products missed, (b) revealed exposure pathways linked to travel distances and service access, and (c) highlighted social vulnerabilities and coping practises that inform equitable intervention priorities. Together, the integrated approach yields place-sensitive insights—translating statistical relationships into implementable targets (e.g., green space per capita, building-to-economic activity benchmarks) for targeted greening, shading, and accessibility improvements. We discuss limitations—namely, the 1 km LST resolution and the small participatory sample—and argue that scaling this framework will better guide locally appropriate, equity-centred heat adaptation in megacities.	